

Opportunity Table Programmatic Extension Runbook

Customer-ready explanation for business stakeholders, product owners, and IT/development teams

Environment: Phoenicarix-CI

Dataverse URL: <https://phoenicarix-ci.crm4.dynamics.com>

Solution: ASO.Core / unique name ASOCore

Table: Opportunity / logical name opportunity

Version: v1.0

Date: 2026-05-23

1. Executive summary

We extended the existing Dynamics 365 / Dataverse Opportunity table programmatically instead of manually creating every column through the maker UI. The purpose was to implement the Phase 2 ASO opportunity schema quickly, consistently, and safely inside the ASO.Core solution.

Plain-language explanation: we added structured boxes to Opportunity records so future AI, SAP, Customer Insights, and sales-orchestration components have a governed place to store their information. No automation was activated yet.

2. Why we used a programmatic approach

- Manual UI creation would require repetitive clicks for every field and increase the risk of wrong data types or names.
- A script creates columns consistently and can be reviewed by IT before execution.
- The script checks whether a field already exists and skips it safely.
- The script associates the new metadata with ASO.Core, preserving solution layering.
- The approach is faster and more repeatable than clicking through dozens of fields manually.

3. Scope of the change

Area	Decision / value
Environment	Phoenicarix-CI
Solution	ASO.Core
Target table	Existing Microsoft Dataverse Opportunity table
Logical table name	opportunity
Publisher prefix	aso
Script technology	.NET 8 console application using Microsoft.PowerPlatform.Dataverse.Client
Change type	Metadata creation: custom columns only
Not included	No flows, no sample data, no SAP connection, no Customer Insights journey, no customer communication

4. Field inventory created on Opportunity

Display name	Schema name	Type	Size / values	Purpose
SAP Customer ID	aso_sapcustomerid	Text	100	SAP customer reference only. Used later when SAP integration is available; not a seller-owned ERP truth.
SAP Order Reference	aso_saporderreference	Text	100	Holds an approved SAP order/commercial reference

				only after governed SAP process.
AI Opportunity Tier	aso_aiopportunitytier	Local choice	A, B, C	Prioritization tier for sales focus.
AI Summary	aso_aisummary	Multiline text	4000	Readable AI-generated account/opportunity summary.
AI Risks	aso_airisks	Multiline text	4000	Risk observations generated by AI or orchestration.
AI Stakeholder Gaps	aso_aistakeholdergaps	Multiline text	4000	Identifies missing buyer personas, champions, decision makers, or blockers.
AI Next Actions	aso_airectactions	Multiline text	4000	Recommended next actions for the seller.
AI Risk Level	aso_airisklevel	Local choice	Low, Medium, High	Simple risk classification for views, filters, and escalation.
AI Follow-up Message Draft	aso_aifollowupmessage	Multiline text	4000	Draft only. It must not be sent directly to customers from Power Automate or Outlook.
AI Confidence	aso_aiconfidence	Decimal	0.00-1.00, precision 2	Confidence score for AI/orchestration output quality.
AI Last Run On	aso_ailastrunon	Date and time	User local	Timestamp of the last AI/orchestration run.
AI Agent Status	aso_aiaгентstatus	Global choice	aso_AIStatus	Reuses the global AI status choice created in Step 2.
AI SAP Commercial Summary	aso_aisapcommercialsummary	Multiline text	4000	Normalized commercial context from SAP after governed reads.
AI SAP Product Summary	aso_aisapproductsummary	Multiline text	4000	Normalized product context from SAP after governed reads.
AI Correlation ID	aso_aicorrelationid	Text	100	Correlation identifier for traceability across Power Automate, Foundry, Dataverse, and future integration layers.
Sales Opportunity Agent Status	aso_salesopportunityagentstatus	Global choice	aso_AIStatus	Status of seller-facing Sales Opportunity Agent output. We reused AI status for MVP consistency.
Sales Opportunity Agent Importance	aso_salesopportunityagentimportance	Text	100	Kept as text because native output may vary by tenant.
Sales Opportunity Agent Risk	aso_salesopportunityagentrisk	Text	100	Kept as text because native output may vary by tenant.
Sales Opportunity Agent Recommendation	aso_salesopportunityagentrecommendation	Multiline text	4000	Seller-facing recommendation from Sales Opportunity Agent or mapped orchestration.

Sales Deal Close Agent Guidance	aso_salesdealcloseagentguidance	Multiline text	4000	Close-readiness guidance for seller/manager review.
Sales Deal Close Agent Last Run On	aso_salesdealcloseagentlastrunon	Date and time	User local	Timestamp of latest Sales Deal Close Agent guidance.
Foundry Final Opportunity Decision	aso_foundryfinalopportunitydecision	Multiline text	4000	Final governed orchestration decision. Multiline allows richer rationale in MVP.
Foundry Review Required	aso_foundryreviewrequired	Yes/No	Default No	Human review gate for low-confidence, risky, or commercially sensitive outcomes.
Communication State	aso_communicationstate	Global choice	aso_CommunicationState	Reuses Customer Insights communication governance state.
Lifecycle Communication Stage	aso_lifecyclecommunicationsstage	Global choice	aso_LifecycleCommunicationStage	Reuses lifecycle-stage taxonomy.
Journey Participation Status	aso_journeyparticipationstatus	Global choice	aso_JourneyParticipationStatus	Reuses journey participation state.
Customer Insights Journey ID	aso_customerinsightsjourneyid	Text	100	Stores latest Customer Insights journey identifier.
Customer Insights Journey Name	aso_customerinsightsjourneyname	Text	200	Stores latest Customer Insights journey name.
Customer Insights Last Entry On	aso_customerinsightslastentryon	Date and time	User local	Timestamp of latest journey entry.
Customer Insights Last Interaction On	aso_customerinsightslastinteractionon	Date and time	User local	Timestamp of latest tracked Customer Insights interaction.
Customer Insights Last Interaction Type	aso_customerinsightslastinteractiontype	Local choice	EmailSent, Open, Click, FormSubmit, Reply, Unsubscribe, Bounce, CustomAction	Local choice for the latest interaction type.
Email Consent Status	aso_emailconsentstatus	Global choice	aso_EmailConsentStatus	Reuses consent status taxonomy.
Compliance Profile Name	aso_complianceprofilename	Text	200	Stores the compliance profile used by Customer Insights.
Communication Hold Reason	aso_communicationholdreason	Multiline text	4000	Explains why communication is blocked, held, or suppressed.

5. Terminal commands used

```
mkdir -p ~/aso-dataverse-tools/opportunity-extension

cd ~/aso-dataverse-tools/opportunity-extension

dotnet new console --framework net8.0

dotnet add package Microsoft.PowerPlatform.Dataaverse.Client

open -a "Visual Studio Code" Program.cs

dotnet run
```

6. How the script worked

Connect to Dataverse: The script opened an OAuth connection to Phoenicarix-CI using the Dataverse client package.

Target the solution: The script used SolutionUniqueName = ASOCore so the metadata was created inside the correct solution layer.

Target the table: The script used EntityLogicalName = opportunity to extend the existing Opportunity table, not create a new table.

Read existing metadata: The script retrieved existing attributes so it could skip already-created fields.

Create missing columns: The script looped over the planned field list and sent CreateAttributeRequest for each missing column.

Handle errors safely: Every field creation was wrapped in try/catch so failures were printed without hiding the exact issue.

Publish metadata: The script published the Opportunity table customizations after successful creation.

7. Validation result

The terminal showed green CREATED messages for the Opportunity fields and ended with a publishing success message. In Power Apps, the ASO.Core solution showed the Opportunity table with the new aso_ columns visible under Columns. This confirms the programmatic extension succeeded.

Validation path: make.powerapps.com -> Phoenicarix-CI -> Solutions -> ASO.Core -> Tables -> Opportunity / Verkaufschance -> Columns -> search aso_

8. Backup naming

After validation, export ASO.Core as both unmanaged and managed. Use these names:

ASO.Core_Phase2_CI_OpportunityExtension_unmanaged_v1_0_20260523.zip

ASO.Core_Phase2_CI_OpportunityExtension_managed_v1_0_20260523.zip

9. Audience-specific explanation

For a non-technical reader

We added extra labelled boxes to the Opportunity record. Later, AI and integration tools can fill those boxes with summaries, risk signals, recommendations, customer journey status, and SAP references.

For the product owner

The Opportunity table is now prepared for AI-assisted opportunity management, risk review, next-best-action guidance, customer communication governance, and future SAP-aware context. No automated decisioning or customer send is live yet.

For the developer / IT team

This is a Dataverse metadata deployment using the SDK. It creates attributes through `CreateAttributeRequest`, uses the ASOCore solution unique name, references global choices where appropriate, creates local option sets where values are table-specific, and publishes opportunity metadata.

10. Governance and safety controls

- Seller-facing AI fields are informative outputs, not automatically executed actions.
- AI Follow-up Message Draft is explicitly draft-only and must not be sent directly from Power Automate or Outlook.
- SAP reference fields are placeholders for governed SAP integration through APIM and Azure Functions; they are not direct SAP writes.
- Foundry Review Required is a human gate for low-confidence or sensitive outcomes.
- Future security design should restrict manual edits on status, confidence, correlation, SAP reference, and final-decision fields.

11. Troubleshooting

Symptom	Likely cause	Action
Connection failed	Wrong Dataverse URL or authentication issue	Authenticate again and verify Phoenicarix-CI URL.
Global choice error	Internal global choice name does not match	Open ASO.Core -> Choices and confirm logical names.
Column already exists	Script was rerun or column was manually created	Safe; script skips existing attributes.
Fields not in ASO.Core	Wrong solution unique name	Verify unique name in pac solution list; script should use ASOCore.
Build error / no Main method	Program.cs content was not pasted correctly	Replace Program.cs with full script and save.

12. Next steps

1. Export managed and unmanaged backups of ASO.Core after Opportunity extension.
2. Continue with Account table extension.
3. Continue with Contact table extension.
4. Later add the new fields to Opportunity forms under AI & Deal Intelligence.
5. Later create operational custom tables such as Agent Run Log and External Sync Ledger.

Appendix A - Script excerpt used for Opportunity

```
using Microsoft.Crm.Sdk.Messages;  
  
using Microsoft.PowerPlatform.Dataverse.Client;  
  
using Microsoft.Xrm.Sdk;  
  
using Microsoft.Xrm.Sdk.Messages;  
  
using Microsoft.Xrm.Sdk.Metadata;
```

```

const string DataverseUrl = "https://phoenicarix-ci.crm4.dynamics.com";

const string SolutionUniqueName = "ASOCore";

const string EntityLogicalName = "opportunity";

const int LanguageCode = 1033;

var connectionString =
    $"AuthType=OAuth;

    Url={DataverseUrl};

    LoginPrompt=Auto;

    ClientId=51f81489-12ee-4a9e-aaae-a2591f45987d;

    RedirectUri=http://localhost";

using var service = new ServiceClient(connectionString);

if (!service.IsReady)
{
    Console.ForegroundColor = ConsoleColor.Red;

    Console.WriteLine("Connection failed:");

    Console.WriteLine(service.LastError);

    Console.ResetColor();

    return;
}

Console.WriteLine($"Connected to: {DataverseUrl}");

Console.WriteLine($"Target solution: {SolutionUniqueName}");

Console.WriteLine($"Target table: {EntityLogicalName}");

Console.WriteLine();

var existingAttributes = GetExistingAttributeLogicalNames(service, EntityLogicalName);

var fields = new List<AttributeMetadata>
{
    Text("aso_sapcustomerid", "SAP Customer ID", 100, "SAP reference only."),
    Text("aso_saporderreference", "SAP Order Reference", 100, "Persist only after approved SAP process."),
    LocalChoice("aso_aiopportunitytier", "AI Opportunity Tier", new[] { "A", "B", "C" }, "AI opportunity tier."),
    Memo("aso_aisummary", "AI Summary", "AI account/opportunity summary."),

```

```

Memo("aso_airisks", "AI Risks", "Identified risks."),

Memo("aso_aistakeholdergaps", "AI Stakeholder Gaps", "Stakeholder coverage gaps."),

Memo("aso_anextactions", "AI Next Actions", "Recommended next actions."),

LocalChoice("aso_airisklevel", "AI Risk Level", new[] { "Low", "Medium", "High" }, "AI risk level."),

Memo("aso_aifollowupmessage", "AI Follow-up Message Draft", "Draft only; no direct customer send."),

DecimalNumber("aso_aiconfidence", "AI Confidence", 0m, 1m, 2, "AI confidence 0.00-1.00."),

DateTimeField("aso_ailastrunon", "AI Last Run On", "Last run timestamp."),

GlobalChoice("aso_aiaagentstatus", "AI Agent Status", "aso_aistatus", "Uses global AI status choice."),

Memo("aso_aisapcommercialsummary", "AI SAP Commercial Summary", "Normalized SAP summary after governed reads."),

Memo("aso_aisapproductsummary", "AI SAP Product Summary", "Normalized SAP product context."),

Text("aso_aicorrelationid", "AI Correlation ID", 100, "Last run correlation."),

GlobalChoice("aso_salesopportunityagentstatus", "Sales Opportunity Agent Status", "aso_aistatus", "Status of seller-facing opportunity agent output."),

Text("aso_salesopportunityagentimportance", "Sales Opportunity Agent Importance", 100, "Native output varies by tenant."),

Text("aso_salesopportunityagentrisk", "Sales Opportunity Agent Risk", 100, "Native output varies by tenant."),

Memo("aso_salesopportunityagentrecommendation", "Sales Opportunity Agent Recommendation", "Seller-facing recommendation."),

Memo("aso_salesdealcloseagentguidance", "Sales Deal Close Agent Guidance", "Close-readiness guidance."),

DateTimeField("aso_salesdealcloseagentlastrunon", "Sales Deal Close Agent Last Run On", "Timestamp."),

Memo("aso_foundryfinalopportunitydecision", "Foundry Final Opportunity Decision", "Final governed orchestration decision."),

YesNo("aso_foundryreviewrequired", "Foundry Review Required", "Human review gate."),

GlobalChoice("aso_communicationstate", "Communication State", "aso_communicationstate", "Uses global Communication State choice."),

GlobalChoice("aso_lifecyclecommunicationstage", "Lifecycle Communication Stage", "aso_lifecyclecommunicationstage", "Uses global Lifecycle Communication Stage choice."),

GlobalChoice("aso_journeyparticipationstatus", "Journey Participation Status", "aso_journeyparticipationstatus", "Uses global Journey Participation Status choice."),

Text("aso_customerinsightsjourneyid", "Customer Insights Journey ID", 100, "Latest journey reference."),

Text("aso_customerinsightsjourneyname", "Customer Insights Journey Name", 200, "Latest journey name."),

DateTimeField("aso_customerinsightslastentryon", "Customer Insights Last Entry On", "Last entry."),

DateTimeField("aso_customerinsightslastinteractionon", "Customer Insights Last Interaction On", "Last interaction."),

LocalChoice("aso_customerinsightslastinteractiontype", "Customer Insights Last Interaction Type", new[] { "EmailSent", "Open", "Click", "FormSubmit", "Reply" }, "Customer Insights Last Interaction Type choice."),

GlobalChoice("aso_emailconsentstatus", "Email Consent Status", "aso_emailconsentstatus", "Uses global Email Consent Status choice."),

Text("aso_complianceprofilename", "Compliance Profile Name", 200, "Profile used."),

Memo("aso_communicationholdreason", "Communication Hold Reason", "Suppression / hold reason.")
};

```



```

foreach (var field in fields)
{
    var logicalName = field.SchemaName!.ToLowerInvariant();
    if (existingAttributes.Contains(logicalName))
    {
        Console.ForegroundColor = ConsoleColor.Yellow;
        Console.WriteLine($"SKIP existing: {logicalName}");
        Console.ResetColor();
        continue;
    }

    try
    {
        Console.WriteLine($"Creating: {logicalName} ...");
        var request = new CreateAttributeRequest
        {
            EntityName = EntityLogicalName,
            Attribute = field,
            SolutionUniqueName = SolutionUniqueName
        };
        service.Execute(request);
        Console.ForegroundColor = ConsoleColor.Green;
        Console.WriteLine($"CREATED: {logicalName}");
        Console.ResetColor();
    }
    catch (Exception ex)
    {
        Console.ForegroundColor = ConsoleColor.Red;
        Console.WriteLine($"FAILED: {logicalName}");
        Console.WriteLine(ex.Message);
        Console.ResetColor();
    }
}

Console.WriteLine();
Console.WriteLine("Publishing Opportunity table customizations...");
service.Execute(new PublishXmlRequest

```

```
{  
    ParameterXml = "<importexportxml><entities><entity>opportunity</entity></entities></importexportxml>"  
});  
Console.ForegroundColor = ConsoleColor.Green;  
Console.WriteLine("Done. Validate in ASO.Core → Tables → Opportunity → Columns.");  
Console.ResetColor();  
  
// Helper functions create the correct Dataverse metadata object for each column type.  
// They keep the main field list readable and prevent copy/paste mistakes.
```